

Education

University of Utah

B.S. Computer Science

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Salt Lake City, UT

August 2020 – May 2025

Skills

Languages: Python, TypeScript, C++ , SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

- L3Harris Technologies

Associate Software Engineer

Dallas, TX

June 2025 – Present

 - Developed signal processing modules in Python and C++ on embedded systems, collaborating with hardware teams to optimize performance and resource utilization.
 - Built microservices for high-frequency data ingestion, processing, and storage, leveraging Podman and serverless functions.
 - Automated CI/CD testing with pipelines, integrating static code analysis and integration testing.
- Doxy.me

Software Engineer

Charleston, SC

August 2024 – May 2025

 - Built a real-time mental health inference system for Doxy.me’s telehealth platform, fine-tuning Llama models to predict depression risk, anxiety, and mood indicators from tone, voice, and facial cues captured over live WebRTC sessions, served via SageMaker and Fargate.
 - Improved prediction accuracy of the mental health inference system by applying RL feedback loops over structured clinical signals and physician-validated outcomes from live video consultations.
 - Shortened the gap between model output and clinician action by developing TypeScript frontend components embedded within the video consultation interface, surfacing AI-generated insights directly at the point of care.
 - Eliminated environment drift and enabled zero-downtime releases by standardizing CI/CD configuration with YAML-based pipelines across the ML stack.
- University of Utah

Undergraduate Research Assistant – FuTURES Lab

Salt Lake City, UT

May 2024 – Jun 2025

 - Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
 - Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.
- HEXstream

Software Engineering Intern

Chicago, IL

May 2022 – Aug 2022

 - Engineered backend ETL pipelines integrating data from 25+ enterprise sources into Azure SQL and Azure Data Lake.
 - Developed automated workflows for ingestion, cleansing, and aggregation, supporting distributed analytics systems.

Projects

University Course Planning/Review Platform (Class Best):

Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Telehealth Mental Health Diagnosis Tool (Doxy.me):

Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

Configuration Fuzzer:

Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.